Question 1: Study Design Selection

A public health team wants to investigate whether long-term exposure to wildfire smoke increases the risk of developing chronic obstructive pulmonary disease (COPD). They identify a population without COPD in 2026, classify participants according to wildfire smoke exposure, and follow them for 15 years to observe incident COPD cases.

a) What study design is being used?
b) What measure of association would be most appropriate?
c) Name one major advantage and one limitation of this design.

Question 2: Calculating and Interpreting Relative Risk

A cohort study follows 8,000 adults for 10 years.

Smoking Status	Develop CHD	No CHD
Smokers	120	3,880
Non-smokers	80	3,920

a) Calculate the incidence of CHD among smokers and non-smokers.
b) Calculate the Relative Risk.
c) Interpret the result.

Question 3: Bias and Validity

A survey examining physical activity among Saskatchewan adults recruits participants through social media advertisements. Individuals who respond tend to be younger and more health-conscious than the general population.

a) What type of bias is most likely present?
b) Does this primarily threaten internal validity or external validity?
c) Briefly explain why.

Answers

Question 1: Answer:

- a) Prospective (concurrent) cohort study.
- b) Relative Risk (RR).
- c) Advantage: establishes temporality and can calculate incidence. Limitation: expensive and may require long follow-up with potential loss to follow-up.

Question 2: Answer:

- Incidence among smokers = 120/4000 = 0.03 (3%)
- Incidence among non-smokers = 80/4000 = 0.02 (2%)
- RR = 0.03 / 0.02 = 1.5
- Interpretation: Smokers have 1.5 times the risk of developing CHD compared with non-smokers. An RR > 1 indicates a positive association between exposure and disease.

Question 3: Answer:

- a) Selection bias.
- b) External validity (generalizability).
- c) The study sample may not be representative of the target population, limiting the ability to generalize findings to all Saskatchewan adults. Selection bias can occur when participant selection is related to study characteristics.